

Trade Liberalization, Intermediate Input Substitution, and the Labor Share*

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June 11, 2026

Abstract

We identify a new channel through which trade liberalization shapes the labor share: intermediate-input substitution under markups. Because value added embeds profits on intermediate inputs, a change in their cost share moves the labor share even when capital–labor margins are fixed. Using China’s WTO accession, we find input-tariff cuts raised the firm-level labor share; absent them, it would have fallen by one-fifth more over 2001–2006. The cost-share channel alone accounts for roughly half of this effect. The estimated elasticity of substitution between primary and intermediate inputs lies predominantly below one, consistent with the channel’s sign. Country-industry evidence from 43 countries corroborates it.

Keywords: labor share; intermediate-input trade; input substitution; trade liberalization.

JEL: F14, F16, D24, D33, E25.

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1 Introduction

The labor share of income has fallen across most economies over the past three decades (Karabarbounis and Neiman, 2014; Grossman and Oberfield, 2022), sharpening interest in how globalization redistributes income between capital and labor. A large literature asks how trade liberalization shapes the labor share, almost always through two margins: the capital–labor cost share—driven by Heckscher–Ohlin forces, offshoring, bargaining power, and factor-biased technology (Elsby et al., 2013; Rodrik, 1997; Ahsan and Mitra, 2014; Leblebicioğlu and Weinberger, 2021; Kondo et al., 2024; Sun, 2020)—or the markup, which trade-induced competition compresses while also inducing a compositional rise of superstar firms with high markups and low labor shares (Autor et al., 2020; Panon, 2022). Both perspectives, however, focus primarily on final goods or capital goods trade and miss a defining feature of modern globalization: intermediate inputs already accounted for roughly 60 percent of international trade by 2010 (Johnson and Noguera, 2012). Yet whether and how this dominant trade flow shapes labor share dynamics is less understood. Beyond the classic channels emphasized in the literature, this paper identifies and quantifies a third channel: *intermediate input substitution* under markups.

A simple decomposition of the labor share reveals a distinct role for intermediate-input trade. The labor share is the share of labor compensation in *value added*,¹ and decomposes into three determinants: the share of labor in primary costs, the markup, and the share of intermediate inputs in total costs. The first two are well understood; the third, far less studied, opens a new channel through which intermediate trade can move the labor share. The intuition follows directly from the definition of value added as gross output net of intermediate-input costs. Under markups, value added comprises not only primary-input compensation but also profits earned on both primary and intermediate inputs. A rise in the intermediate-input share therefore expands the profits booked on intermediates, inflating value added without raising labor compensation and mechanically lowering the labor share. Input trade liberalization sets this margin in motion: by lowering the relative price of intermediates, it leads firms to substitute intermediates for primary inputs, shifting cost shares and hence the labor share. The direction depends on the elasticity of substitution between the two—a fall in the price of intermediates raises the labor share if primary and intermediate inputs are gross complements (elasticity below one) and lowers it if they are gross substitutes (elasticity above one). We call this the *intermediate input substitution channel*.

We test the channel using China’s accession to the World Trade Organization (WTO) in 2001. Following Mo et al. (2021), we distinguish input, capital, and output tariffs at the four-digit Chinese industry level, and estimate firm-level labor-share responses to industry-level tariff variation. Only the intermediate-input tariff has a stable, statistically significant effect: firms in industries that experienced larger input-tariff cuts raised their labor share relative to firms in industries that experienced smaller cuts. To address endogeneity, we instrument the time-varying input tariff with

¹We measure the labor share in value added rather than in gross output for two reasons. First, the aggregate labor share is defined in terms of value added (GDP), and the value-added measure is standard in the labor-share literature. Second, the gross-output measure double-counts labor, since intermediate inputs are themselves produced by upstream primary factors, including labor.

the pre-WTO (2001) industry tariff interacted with a post-2001 dummy (Goldberg and Pavcnik, 2005; Brandt et al., 2017; Liu et al., 2019). Industries that entered WTO accession with higher protection mechanically experienced the largest subsequent cuts, while 2001 tariff levels were set well before our sample period. The IV estimate is sharper than OLS, a flexible event-study finds no pre-trends, and the result survives a battery of robustness checks reported in the Online Appendix.

The effect is economically large. Between 2001 and 2006 the firm-median labor share fell from 33.6 to 28.8 percent, a drop of 4.86 percentage points. Applying our preferred IV estimate to observed industry-by-year input-tariff cuts, had tariffs remained at their 2001 levels the firm-median labor share would have fallen by an additional 0.96 percentage point—about one-fifth of the total observed decline.

We then decompose this effect through the three determinants of the identity. Input-tariff cuts significantly lower the intermediate-input cost share, the margin through which the intermediate input substitution channel operates. Of the two classic margins, the cuts also compress the markup but leave the labor share of primary costs statistically unchanged. Translating these responses into channel contributions, the intermediate-input substitution channel alone accounts for 48 percent of the total labor-share response, markup compression for 62 percent, and the labor-cost-share channel for a small −13 percent. Nearly half of the labor-share gain from input-tariff cuts therefore operates through a margin that prior accounts of trade and the labor share have largely set aside.

As a structural check on the sign, we estimate the elasticity of substitution between intermediate inputs and primary inputs σ_{HM} from cross-prefecture wage variation in China following Oberfield and Raval (2021). For the vast majority of two-digit manufacturing industries the point estimate lies below one, with a pooled value of $\hat{\sigma}_{HM} \approx 0.85$. This is exactly the sign restriction implied by the reduced-form cost-share response, and it is identified out-of-sample from cross-section price dispersion rather than from the WTO tariff time series.

Finally, we extend our analysis to a comprehensive global country-industry level sample of 43 countries with 56 industries during 2000-2014 to examine the generalizability of our findings. After controlling for industry, country, and year fixed effects, we show that lower intermediate input tariff rates are associated with a significantly higher industry labor share in this global sample, consistent with our findings among Chinese firms.

We contribute to two literatures. First, to the literature on trade and the labor share, we identify intermediate input substitution as a third margin—alongside capital-labor cost share and markup²—and provide the first causal evidence that trade-policy shocks activate it. The accounting decomposition has been used in some recent research on the labor share in general (Giannoni and Mertens, 2019; Castro-Vincenzi and Kleinman, 2022; Hang, 2025), but its link to trade liberalization has not been quantified. Our paper is most closely related to three studies on China’s WTO accession and the labor share. Miao et al. (2023) also find an important role for input tariffs but pool all imported goods into a single tariff and focus on markups and reallocation; following Mo et al. (2021), we instead distinguish input types and isolate intermediate inputs. Kondo et al. (2024) show that input-tariff cuts raise the

²Literature related with these two margins has been discussed in the beginning of the paper. The broader literature on labor share is too large to be summarized here. For overviews please refer to Grossman and Oberfield (2022) and Karabarbounis (2024).

labor share by compressing labor markdowns; we show the channel operates even absent any markdown response. [Kamal et al. \(2019\)](#) also find that tariff cuts raise the labor share, but measure labor’s share of *gross output*, whereas we use the *value-added* share—the relevant object for the aggregate labor share and the one on which the substitution mechanism hinges.

Second, we contribute to the literature on the firm-level effects of input trade liberalization ([Amiti and Konings, 2007](#); [Topalova and Khandelwal, 2011](#); [Halpern et al., 2015](#); [Goldberg et al., 2010](#); [Bas and Strauss-Kahn, 2015](#); [Fan et al., 2015](#)) by linking a distributional outcome to the input-tariff margin and by providing a decomposition that organizes the three forces any explanation must work through.

The remainder of the paper is organized as follows. Section 2 sets out the decomposition and the sign condition. Section 3 describes the data and identification. Section 4 presents the baseline and IV results. Section 5 quantifies the three channels and estimates σ_{HM} . Section 6 presents the cross-country evidence, and Section 7 concludes.

2 Theoretical Framework

Decomposition of the labor share. Consider a production unit that uses capital (K), labor (L), and intermediate inputs (M) at prices R , W , and q , with constant returns to scale and a markup $\mu \geq 1$ over total input costs. Gross output is $V = \mu(RK + WL + qM)$, and value added is $Y = V - qM = (RK + WL) + (\mu - 1)(RK + WL) + (\mu - 1)qM$. The last term, profits earned on intermediate inputs, is the object that links intermediate trade to the labor share. Let $\theta_L \equiv WL/(RK + WL)$ be the share of labor in primary costs and $\alpha_M \equiv qM/(RK + WL + qM)$ be the share of intermediate inputs in total costs. Then

$$s_L \equiv \frac{WL}{Y} = \frac{\theta_L}{1 + \frac{\mu - 1}{1 - \alpha_M}}. \quad (1)$$

The labor share is determined by three statistics: the labor intensity of primary production θ_L , the markup μ , and the intermediate-input cost share α_M . The first two are the focus of the prior literature; the third is the new margin we exploit. Under perfect competition ($\mu = 1$) the profit terms vanish and only θ_L matters, so the labor share is determined entirely by the capital–labor mix. Under imperfect competition ($\mu > 1$), however, firms earn economic profits on *both* primary inputs and intermediate inputs, and the profit wedge on intermediate inputs $(\mu - 1)qM$ inflates value added without increasing labor compensation.

A simple numerical illustration clarifies the mechanism. Consider a firm with $\mu = 1.5$ that uses labor as the only primary input. With \$100 of labor and zero intermediates, value added is \$150 (\$100 of labor cost plus \$50 of profit on labor) and the labor share is 2/3. Adding \$100 of intermediates raises gross output to \$300; value added becomes \$200, comprising \$100 of labor cost, \$50 of profit on labor, and an additional \$50 of profit generated by marking up the intermediates. The labor share falls to 1/2, solely because intermediates enter the cost base under markups. The derivation of (1) from $Y = \mu(RK + WL + qM) - qM$ is reproduced in Appendix D.

Totally differentiating (1) yields the decomposition in changes:

$$\hat{s}_L = \hat{\theta}_L - \left(\frac{V}{Y}\right)\hat{\mu} + \left(\frac{\Pi}{Y}\right)(1 - \alpha_M), \quad (2)$$

where $\hat{x} \equiv dx/x$ and $\Pi = Y - (RK + WL)$. The three terms map directly to existing strands of the literature (capital-labor margin, markup channel) and to the new intermediate-input substitution channel proposed here.

Sign of the intermediate-input cost-share response. How intermediate-input tariff cuts affect the two classic margins is well understood; how they affect the intermediate-input substitution margin is not. A cut in the intermediate-input tariff lowers the relative price of intermediate inputs, leading firms to use more intermediates relative to primary inputs. Whether the intermediate-input cost share α_M rises or falls, however, depends on the elasticity of substitution between primary and intermediate inputs, σ_{HM} . When $\sigma_{HM} > 1$ (gross substitutes), a 1 percent fall in the relative price raises the relative quantity by more than 1 percent, so spending on intermediates rises relative to primary inputs: α_M increases and the labor share falls. When $\sigma_{HM} < 1$ (gross complements), the quantity response is muted, the price effect dominates, and α_M falls—raising the firm’s labor share. We thus obtain:

Proposition 1 *An input-tariff reduction raises (reduces) the firm labor share through the intermediate input substitution channel if and only if primary and intermediate inputs are gross complements, $\sigma_{HM} < 1$ ($\sigma_{HM} > 1$).*

Proposition 1 provides both a sign prediction for our reduced-form estimates and a structural moment ($\sigma_{HM} \leq 1$) that we independently test in the data. The markup and labor-share-of-primary-costs channels operate in addition, so we estimate all three in Section 5.

3 Data and Empirical Strategy

We exploit the large, sudden reductions in import tariffs across industries following China’s accession to the WTO for identification, a strategy widely used in the literature (e.g., Brandt et al., 2017). We estimate the impact of intermediate-input tariffs on the firm-level labor share using

$$y_{ijt} = \alpha + \beta \log(1 + \tau_{jt}^M) + \Gamma' X_{ijt} + \gamma_i + \gamma_t + \epsilon_{ijt}, \quad (3)$$

where i indexes a firm, j a four-digit Chinese Industrial Classification (CIC) industry, and t a year between 2000 and 2006. The outcome y_{ijt} is the log labor share (denoted ls_{it} in the tables that follow), defined as labor compensation (wages plus fringe benefits including paid leave and employer contributions to pension and health insurance) divided by value added, where value added is gross output net of intermediate-input expenditure. Following Mo et al. (2021), τ_{jt}^M is the average tariff on industry j ’s intermediate inputs in year t , weighted by the input’s import-share from Customs data. The inputs are classified as intermediates using the United Nations Broad Economic Categories (BEC). The vector X controls for the capital import tariff τ_{jt}^K , defined analogously to τ_{jt}^M but over capital goods,

and the output tariff τ_{jt}^Q , the import tariff on industry j 's output in year t . Firm and year fixed effects, γ_i, γ_t , absorb time-invariant firm characteristics and aggregate shocks. Standard errors are clustered at the four-digit industry level.

Identification. A concern with OLS is that tariff schedules may be set strategically. Policy makers may grant smaller tariff cuts to industries with less competitive domestic incumbents, or industries with declining labor shares may lobby for slower liberalization. If protected industries also have unobservable downward trends in the labor share, OLS conflates political economy with the proposed mechanism. We address this with an instrumental variable that exploits China's WTO accession in 2001, an approach widely used in the trade-and-China literature (Brandt et al., 2017; Liu et al., 2019). Following Goldberg and Pavcnik (2005), we instrument $\log(1 + \tau_{jt}^M)$ with the 2001 pre-WTO tariff $\log(1 + \tau_{j,01}^M)$ interacted with a post-2001 indicator. WTO accession required convergence to uniform multilateral standards, so industries that entered with higher protection mechanically experienced larger subsequent cuts; the 2001 tariff levels themselves were pre-determined years before our sample period as part of multilateral negotiations and were stable in the immediate pre-WTO years, making them plausibly exogenous to industry-level dynamics that drive contemporaneous labor-share outcomes. The exclusion restriction requires that the initial level of protection, interacted with the post-2001 indicator, affects the labor share only through subsequent tariff cuts. We perform two main tests for the identifying assumption: (i) a flexible event-study version of (3) that interacts the 2001 tariff with year dummies, reported in Column (5) of Table 1, and (ii) explicit controls for concurrent policy reforms (trade policy uncertainty, export-market access, SOE restructuring, FDI liberalization), reported in Appendix A. The event-study coefficients on the pre-WTO years are statistically insignificant and small in magnitude, while the post-WTO coefficients are positive, significant, and growing over time—a pattern that supports parallel trends and is hard to reconcile with selection on differential pre-trends.

Data. The firm-level data are the Chinese Annual Survey of Industrial Firms (ASIF), compiled by the National Bureau of Statistics. It covers all state-owned enterprises and non-state firms above an annual-sales threshold of five million RMB (roughly USD 800,000), spans 1998–2007 (with substantial improvements in coverage over time, from around 140,000 firms in the late 1990s to over 330,000 by 2007), and contains firm identifiers and detailed operational and accounting information (Brandt et al., 2017). We retain manufacturing firms over 2000–2006 and impose standard cleaning rules: we drop firms that violate basic accounting identities (such as fixed assets or liquid assets exceeding total assets), firms with fewer than eight workers (which are subject to a different legal regime in China), firms with implausible founding dates (such as 1900 or earlier), firms with non-positive wage bill or intermediate-input expenditure, and firms whose reported intermediate-input expenditure exceeds gross output (which would imply negative value added and is a likely sign of data error). Labor compensation combines direct wage payments with fringe benefits including paid leave, commissions, and employer contributions to pension plans and health insurance, which constitute a substantial share of total labor cost in the Chinese context, especially among state-owned and large private enterprises (Rickne, 2013). Merging with the Mo et al. (2021) industry tariffs yields an unbalanced

panel of approximately 1.2 million firm-years. Appendix Table A.1 reports summary statistics.

Measuring θ_L , μ , and $1-\alpha_M$. The mechanism regressions in Section 5 replace the labor share with $1-\alpha_M$, μ , or θ_L on the left-hand side, so we first describe how these three objects are constructed. The data report wage payments and intermediate-input expenditure directly, but not the markup or capital costs. Capital cost is the product of capital stock and rental price. We obtain firm’s real capital stock from Brandt et al. (2012) (perpetual-inventory method), with a depreciation rate of 9 percent. The rental rate of capital is computed as in Oberfield and Raval (2021), $R_{jt} = [P_{j,t-1}r_t + \delta_{jt}P_{jt} - (P_{jt} - P_{j,t-1})]/(1 - tax_t)$, where r_t is the real interest rate (World Bank), P_{jt} is the NBS industry-level capital price index, and tax_t is the effective corporate income tax rate—25 percent for domestic firms and 15 percent for foreign-invested firms, following Kong et al. (2008), which is lower than the statutory rates that prevailed before the 2008 reform. With capital costs in hand, the firm-level markup is constructed as the accounting ratio of total revenue to total costs, which avoids the well-known identification and data issues of the production-function approach to markup estimation (Bond et al., 2021). The intermediate-input cost share α_M is then the ratio of intermediate-input expenditure to total cost. The labor share of primary costs θ_L is the ratio of the wage bill to total primary-input cost (capital plus labor), $WL/(RK + WL)$.

4 Baseline and IV Results

Descriptive evidence. Before turning to formal estimation, we examine the raw data. Splitting four-digit CIC industries at the median of the 2001–2006 reduction in τ_{jt}^M , the mean firm-level labor share in the high-cut (“treated”) group and the low-cut (“control”) group track each other closely in 2000 and 2001—the two groups stand within about half a percentage point of one another and exhibit similar pre-WTO trends—but begin to diverge from 2002 onward (See Appendix Figure A.1). By 2006, firms in industries with above-median input-tariff cuts have a labor share roughly five percentage points higher than firms in industries with below-median cuts. Both groups display the secular decline in the labor share documented for Chinese manufacturing more broadly, but the decline is markedly attenuated for the high-exposure group. The pattern is consistent with input trade liberalization acting as a buffer against the secular forces driving the decline in the labor share.

Baseline. Table 1 reports the central regression. Column (1) gives the OLS estimate of β in (3) from our preferred specification, which controls for the capital and output tariffs in addition to firm and year fixed effects. The coefficient is negative and significant.³

IV estimation. Columns (2)–(4) instrument $\log(1 + \tau_{jt}^M)$ with the 2001 tariff level interacted with a post-2001 dummy. Column (2) reports the first stage results confirming that industries with higher pre-WTO tariffs experienced systematically larger subsequent cuts. The reduced form in Column (3) indicate that firms in industries with higher initial protection—and therefore larger mandated tariff cuts—experienced a relative rise in the labor share after WTO accession. The IV estimate in Column (4) is negative and significant. Its magnitude is larger than the OLS

³Appendix Table A.2 shows that the OLS baseline is stable as the controls are added step by step.

estimate, consistent with classical measurement error in the time-varying tariff attenuating the OLS coefficient and with the IV isolating the WTO-mandated component of the tariff cuts. Column (5) reports the flexible event-study that interacts the 2001 input tariff with year dummies (2002 omitted). The 2000 and 2001 interactions are small and statistically insignificant, indicating no differential pre-trends before WTO accession, while the post-2001 coefficients are positive, significant, and grow monotonically from 0.377 in 2003 to 1.337 in 2006. This dynamic pattern supports the parallel-trends assumption underlying the instrument and shows that the labor-share gains materialized only after the tariff cuts took effect.

Economic magnitude. The IV estimate is also economically meaningful. Between 2001 and 2006 the firm-median manufacturing labor share fell from 33.6 to 28.8 percent, a 4.86 percentage-point decline. Applying the IV coefficient $\hat{\beta} = -0.688$ to the realized industry-by-year input-tariff cuts $\Delta \log(1 + \tau_{jt}^M)$ implies that, had tariffs remained at their 2001 levels, the firm-median labor share would have fallen by an additional 0.96 percentage point—about one-fifth of the total decline. As Section 5 shows, the intermediate-input substitution channel alone accounts for roughly half of this counterfactual gain.

Robustness. Appendix A reports a battery of robustness checks. First, the estimate is stable to controlling for concurrent policy reforms that coincided with WTO entry: trade policy uncertainty (the interaction of pre-WTO column-2 tariffs with a post-2001 indicator, capturing the reduction in trade policy uncertainty highlighted by Handley and Limão, 2017), simultaneous improvements in export-market access (the industry-level export tariff, capturing reciprocal tariff reductions by China’s trading partners), SOE reform (industry SOE share interacted with year dummies, absorbing privatization and restructuring), and the 2002 FDI liberalization (an indicator for industries newly subject to relaxed FDI entry restrictions interacted with a post-2002 dummy). Second, the estimate is unchanged on a balanced panel of continuously observed firms, after adding prefecture-by-year fixed effects that absorb regional shocks such as the “Go West” development program and minimum-wage changes, after winsorizing the top and bottom 5 percent of the labor-share distribution, and after adding the consumption-goods tariff. Third, replacing the import-share-weighted input tariff with an alternative measure that uses China’s 2002 input–output coefficients as weights also leaves the result intact, addressing concerns about how within-industry imports are aggregated. Across all these specifications, the IV coefficient remains negative and significant.

5 Mechanisms

The decomposition in (2) attributes the labor-share response to three margins. We now (i) estimate how each margin responds to the input tariff, (ii) translate those estimates into the contribution of each channel to total labor-share response, and (iii) provide an out-of-sample structural check on the sign of the intermediate-input cost-share response.

5.1 Effect on the Determinants of the Labor Share

Table 2 replaces the labor share on the left-hand side of the IV regression with $1 - \alpha_M$, μ , and θ_L in turn; Column (1) reproduces the labor-share IV for reference.

The point estimate on $1 - \alpha_M$ in Column (2) is -0.640 and significant at the 5 percent level. A reduction in the input tariff therefore *raises* the share of primary inputs in total costs and, equivalently, *lowers* the share of intermediate inputs α_M . As cheaper imported intermediates lead firms to spend relatively less on intermediates, the profit wedge that markups generate on those intermediates shrinks relative to value added, leaving a larger share of value added for primary inputs and lifting the labor share. This is exactly the sign predicted by Proposition 1 when primary and intermediate inputs are gross complements ($\sigma_{HM} < 1$); we confirm this sign restriction directly in Section 5.3 using a different source of variation.

The markup coefficient in Column (3) is positive and significant, indicating that lower input tariffs depress markups. Although cheaper inputs lower marginal cost and can raise markups in partial equilibrium, the pro-competitive general-equilibrium force—entry and intensified competition in models such as Melitz and Ottaviano (2008)—dominates; Appendix D.3 formalizes this result with imported intermediates.

The coefficient on θ_L in Column (4) is small and statistically insignificant, so input-tariff cuts do not measurably shift the labor share of primary costs. The estimate bundles capital–labor substitution with any labor-markdown response (Kondo et al., 2024); its small magnitude implies that the bulk of the labor-share response operates through the cost-share and markup margins.

5.2 Quantification of the Three Channels

To translate the determinant regressions into the contribution of each channel to the total labor-share response, we combine the IV coefficients in Table 2 with the sample medians of V/Y and Π/Y in (2). Figure 1 reports the three contributions: the intermediate-input substitution channel ($\widehat{1 - \alpha_M}$) accounts for 48 percent of the total response, the markup channel ($\hat{\mu}$) for 62 percent, and the labor-cost-share channel ($\hat{\theta}_L$) for -13 percent.⁴

The intermediate-input substitution channel is large in magnitude: nearly half (48.3 percent) of the labor-share gain from input liberalization operates through a margin that prior work on trade and the labor share has not isolated. Ignoring it yields an incomplete account of how trade liberalization shapes the labor share.⁵

The size of the markup channel reflects both a large response of the markup to input tariffs and a particularly large ratio of gross output to value added in the data. In our sample, intermediates account for roughly 70 percent of gross output, implying V/Y above three (against around two for the United States, Jones, 2011). Since (2) scales the markup coefficient by V/Y , the same markup response translates into a quantitatively larger contribution in an economy with deeper input–output linkages, which is why the markup margin features so prominently here despite

⁴The three terms sum to 97 percent rather than exactly 100 because the decomposition is in changes (finite, not infinitesimal) and because the median ratios are computed independently across firms.

⁵Appendix Table B.1 shows that across two-digit CIC industries the channel accounts for 40–68 percent of the total response.

a moderate coefficient on $\hat{\mu}$ in Table 2. The negative -13 percent contribution from $\hat{\theta}_L$ is suggestive rather than definitive, given the insignificant underlying coefficient.

5.3 Elasticity of Substitution between Primary and Intermediate Inputs

To check the sign restriction from Proposition 1 using a different source of variation, we estimate σ_{HM} directly from cross-prefecture wage dispersion in China, following Oberfield and Raval (2021). Under the nested CES production function from Section 2, the first-order condition implies

$$\log(c_H H / q M) = (1 - \sigma_{HM}) \log(q / c_H) + \text{constant}. \quad (4)$$

Assuming intermediates and capital have the same prices across regions, regional wage differences—driven in part by *Hukou* migration frictions—trace out the variation in c_H . For each two-digit CIC industry, we run

$$\log[(rK_{ic} + wL_{ic}) / (qM_{ic})] = (1 - \sigma_{HM})(1 - \alpha_{ic}) \log w_c + \gamma' X_{ic} + \epsilon_{ic},$$

where i indexes firms, c prefectures, $\alpha_{ic} = rK_{ic} / (rK_{ic} + wL_{ic})$ scales the wage shock into a shock to the primary-bundle cost, and w_c is the prefecture-level residual wage from a Mincer regression on the 2005 one-per-thousand population census (controlling for age, education, occupation, and demographics; details in Appendix B). X_{ic} includes four-digit industry, ownership, and firm-age controls; standard errors are clustered at the province level. Identification comes from spatial wage dispersion that is, by construction, orthogonal to the WTO time-series variation used in the baseline.

Figure 2 reports the industry-level estimates: they lie predominantly between 0.6 and 1.1, with almost all industries below one and only four above it.⁶ The pooled estimate is $\hat{\sigma}_{HM} \approx 0.85$. These estimates are consistent with the reduced-form intermediate-input cost-share response and confirm gross complementarity, $\sigma_{HM} < 1$.⁷

To summarize the mechanisms, input-tariff cuts lower the intermediate-input cost share and the markup while leaving θ_L unchanged; the decomposition attributes roughly 48 percent of the labor-share response to intermediate-input substitution, with most of the remainder operating through markup compression. The elasticity estimate confirms the sign of the cost-share response from an independent source of variation.

⁶Detailed industry-level estimates are reported in Appendix Table B.3.

⁷This accords with existing estimates of this elasticity, which also point to complementarity: Atalay (2017) estimates 0.8–0.9 for U.S. industries and Peter and Ruane (2023) find gross complementarity for Indian plants. Near-zero estimates such as Boehm et al. (2019) recover a short-run elasticity from supply disruptions, whereas our cross-sectional design, like Oberfield and Raval (2021), identifies a longer-run one.

6 Global Evidence

The mechanism comes from an accounting decomposition of the definition of labor share, so it should also operate outside China. To check this, we estimate (3) at the country–industry–year level using the World Input–Output Database’s Socio-Economic Accounts for 43 countries and 56 manufacturing industries over 2000–2014. Input tariffs are constructed from HS6 WITS tariffs aggregated to ISIC Rev.4 industries using BEA-style intermediate/capital/consumption splits and import-share weights. Country, industry, and year fixed effects are included; we estimate with OLS because pre-determined country-by-industry tariff variation analogous to China’s WTO accession is unavailable globally.

Table 3 reports the results. Consistent with the Chinese firm-level evidence, the intermediate-input tariff has a negative and statistically significant effect on the industry labor share. The capital tariff is insignificant; the output tariff is significant but an order of magnitude smaller. The intermediate-tariff coefficient is also smaller than at the firm level, as expected when aggregation raises σ_{HM} (Oberfield and Raval, 2021); the sign nevertheless remains negative, indicating gross complementarity on average.

7 Conclusion

This paper identifies intermediate input substitution under markups as a channel through which trade liberalization shapes the labor share. Starting from a decomposition of the labor share into θ_L , μ , and α_M , we show that changes in the intermediate cost share move the labor share whenever firms earn markups on total input costs.

Using China’s WTO accession, we find that input-tariff cuts raised the firm-level labor share, cushioning the observed decline by about one-fifth over 2001–2006. Only intermediate-input tariffs matter once capital and output tariffs are controlled for. The intermediate-input substitution channel accounts for 48 percent of the total effect; markup compression absorbs most of the remainder. An independent estimate of σ_{HM} from cross-prefecture wage variation yields $\hat{\sigma}_{HM} \approx 0.85$, consistent with the sign of the cost-share response. Country-industry evidence from 43 countries confirms that the qualitative pattern is not specific to China.

Firm-level and aggregate labor shares need not move in the same direction when σ_{HM} rises with aggregation (Oberfield and Raval, 2021); pinning down the aggregate elasticity is an important agenda for future research. More broadly, any complete account of the global decline in the labor share needs to include the cost share of intermediate inputs alongside the more familiar capital–labor and markup margins. Mapping the firm-level cost-share response to the aggregate factor share through input–output linkages remains a structural question we leave to future work.

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Tables and Figures

Table 1: *The Effect of Input Tariff Reductions on Firm Labor Share.*

	(1)	(2)	(3)	(4)	(5)
	OLS	First-stage	Reduced-form	IV	Flexible
DepVar:	ls_{it}	τ_{jt}^M	ls_{it}	ls_{it}	ls_{it}
$\log(1 + \tau_{jt}^M)$	-0.473** (0.212)			-0.688*** (0.244)	
$\log(1 + \tau_{j,01}^M) \times Post01_t$		-0.309*** (0.045)	0.707*** (0.240)		
$\log(1 + \tau_{j,01}^M) \times Y2000_t$					-0.226 (0.180)
$\log(1 + \tau_{j,01}^M) \times Y2001_t$					-0.116 (0.115)
$\log(1 + \tau_{j,01}^M) \times Y2003_t$					0.377** (0.192)
$\log(1 + \tau_{j,01}^M) \times Y2004_t$					1.109*** (0.261)
$\log(1 + \tau_{j,01}^M) \times Y2005_t$					1.236*** (0.237)
$\log(1 + \tau_{j,01}^M) \times Y2006_t$					1.337*** (0.309)
$\log(1 + \tau_{jt}^K)$ control	Y	Y	Y	Y	Y
$\log(1 + \tau_{jt}^Q)$ control	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Kleibergen-Paap F	—	272.299	—	—	—
R^2	0.588	—	—	—	—
Observations	1,205,220	1,205,220	1,205,220	1,205,220	1,205,220

Notes: Observations are at firm-year level from 2000 to 2006. The dependent variable in Columns (1), (3), (4), and (5) is the logarithm of the firm-level labor share. Column (2) is the first stage with the four-digit industry input tariff on the left-hand side. Column (5) reports the flexible event-study that interacts the 2001 pre-WTO input tariff $\log(1 + \tau_{j,01}^M)$ with year dummies, taking 2002 (the first full year after WTO accession) as the omitted reference year. The pre-WTO interactions for 2000 and 2001 are small and statistically insignificant, while the post-WTO interactions are positive, significant, and grow monotonically through 2006, supporting the parallel-trends assumption. All specifications control for the capital tariff τ_{jt}^K and the output tariff τ_{jt}^Q and include firm and year fixed effects. Standard errors clustered at the industry level. *, **, *** indicate significance at the 10%, 5%, and 1% levels.

Table 2: *The Effect of Input Tariff Reductions on the Determinants of the Labor Share.*

	(1)	(2)	(3)	(4)
	IV	IV	IV	IV
DepVar:	s_L	$1 - \alpha_M$	μ	θ_L
$\log(1 + \tau_{jt}^M)$	-0.688*** (0.244)	-0.640** (0.294)	0.102** (0.043)	0.090 (0.095)
Controls	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Kleibergen-Paap F	272.299	272.299	272.299	272.299
Observations	1,205,220	1,205,220	1,205,220	1,205,220

Notes: All columns are IV regressions; the instrument is the 2001 pre-WTO input tariff interacted with a post-2001 indicator. Controls and fixed effects match Column (4) of Table 1. Standard errors clustered at the industry level. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 3: *The Effect of Input Tariffs on Industry Labor Share: Global Evidence.*

	(1)	(2)	(3)
	OLS	OLS	OLS
DepVar:	ls_{it}	ls_{it}	ls_{it}
$\log(1 + \tau_{jt}^M)$	-0.0153** (0.0077)	-0.0164** (0.0078)	-0.0282** (0.0080)
$\log(1 + \tau_{jt}^K)$		0.0179 (0.0198)	0.0129 (0.0198)
$\log(1 + \tau_{jt}^Q)$			0.0013*** (0.0002)
Country FE	Y	Y	Y
Industry FE	Y	Y	Y
Year FE	Y	Y	Y
R^2	0.459	0.459	0.460
Observations	15,887	15,887	15,887

Notes: Country–industry–year observations from 2000 to 2014 from the WIOD Socio-Economic Accounts (43 countries, 56 manufacturing industries). The dependent variable is the log industry labor share. The intermediate-input tariff is constructed by aggregating HS6 tariff rates from WITS to ISIC Rev.4 industries using the UN concordance, applying BEA intermediate/capital/consumption splits, and weighting by within-industry import shares. Capital and output tariffs are constructed analogously. Standard errors clustered at the industry level. *, **, *** denote significance at the 10%, 5%, and 1% levels.

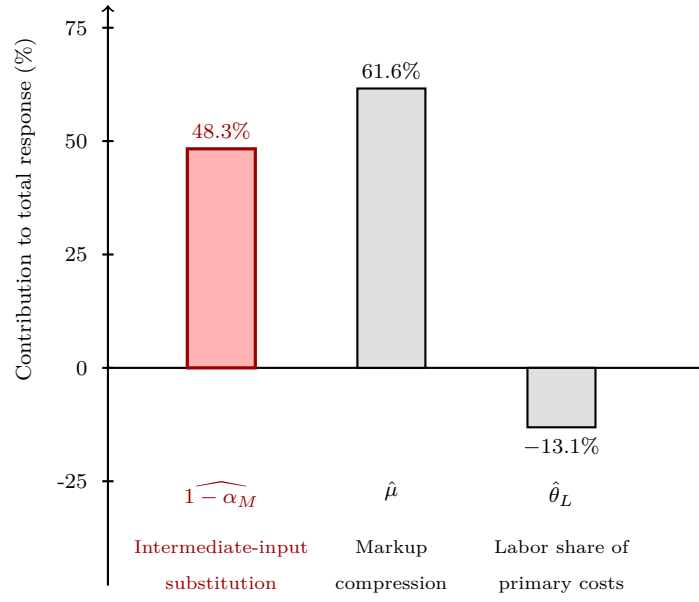


Figure 1: Contribution of Each Channel to the Labor-Share Response to Input-Tariff Cuts

Notes: Each bar reports the percentage of the total labor-share response attributable to one term in (2), computed from the IV estimates in Table 2 using the sample medians of V/Y and Π/Y . The intermediate-input substitution channel ($\widehat{1 - \alpha_M}$, highlighted) accounts for 48 percent of the total response. Appendix Table B.1 reports the same decomposition by two-digit industry and using sample means.

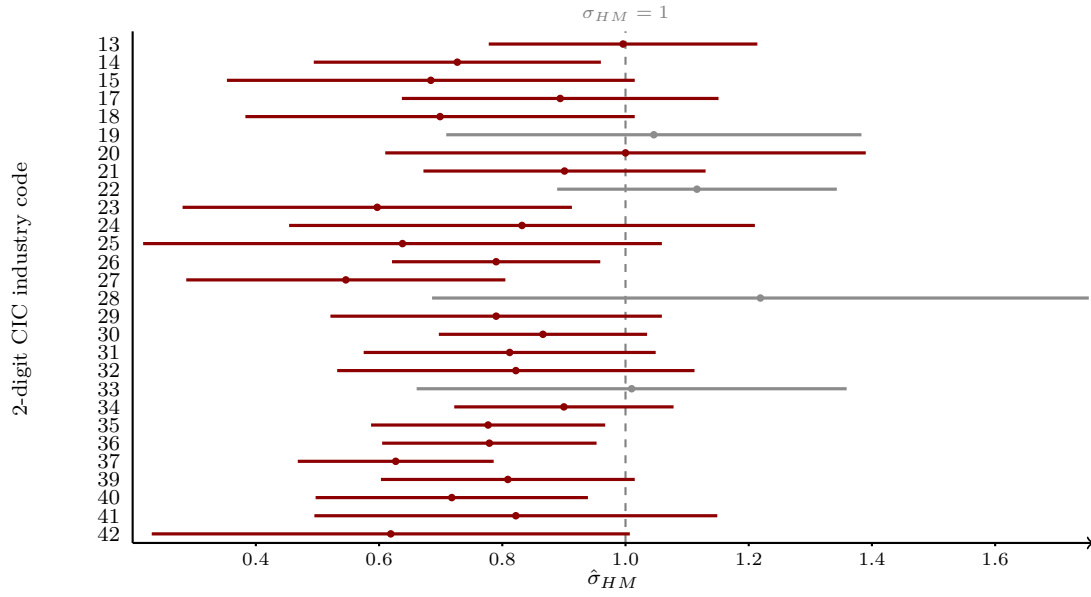


Figure 2: Estimates of the Elasticity of Substitution between Primary and Intermediate Inputs

Notes: Each row plots $\hat{\sigma}_{HM}$ for one two-digit CIC industry, with 95% confidence intervals. Segments are red when $\hat{\sigma}_{HM} \leq 1$ and gray when $\hat{\sigma}_{HM} > 1$. The vertical dashed line marks $\sigma_{HM} = 1$. Estimation follows [Oberfield and Raval \(2021\)](#); detailed point estimates are in Appendix Table B.3.

Online Appendix

A Robustness Checks and Additional Empirical Results

A.1 Graphical Evidence: Labor-Share Dynamics by Tariff-Cut Exposure

To visualize the descriptive evidence discussed in Section 4, we split four-digit CIC industries at the median of the 2001–2006 reduction in τ_{jt}^M and plot the mean firm-level labor share for the high-cut (“Treated”) and low-cut (“Control”) groups over 2000–2006.

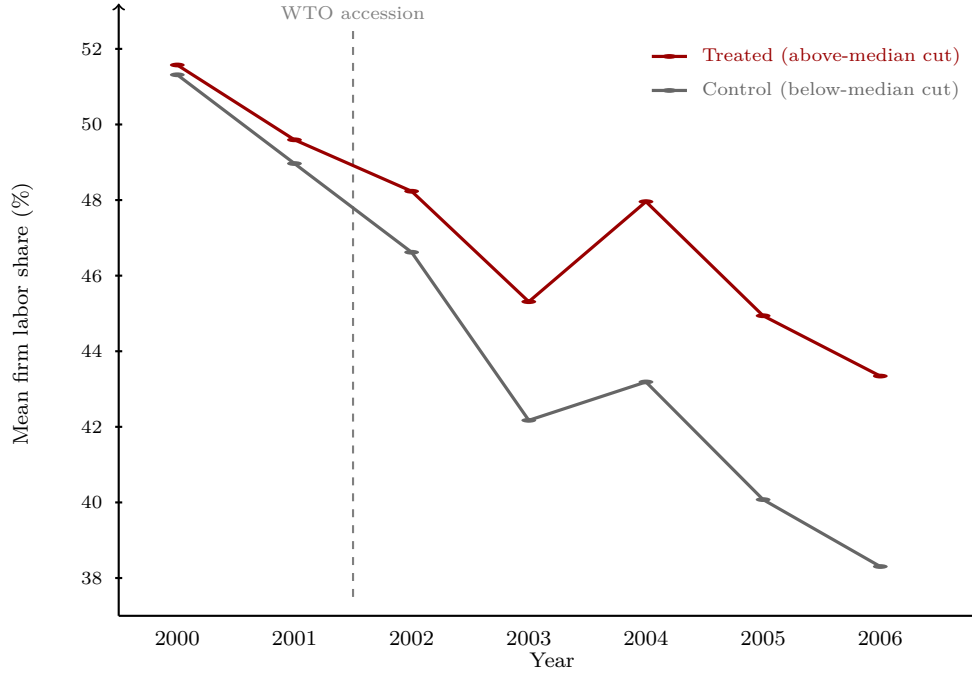


Figure A.1: Labor Share Dynamics During 2000–2006

Notes: Mean firm-level labor share by year, separately for industries with above-median (*Treated*) and below-median (*Control*) reductions in the intermediate-input tariff over 2000–2006. Vertical dashed line marks China’s WTO accession (end of 2001).

A.2 Summary Statistics

Table A.1: *Summary Statistics.*

	Obs.	Mean	Median	Std.dev.	Min	Max
log (Labor share)	1,205,220	−1.278008	−1.171282	1.027875	−4.778411	1.669901
Intermediate input tariff	1,205,220	0.07343	0.0682147	0.0358892	0.0006428	0.2956958
Capital good tariff	1,205,220	0.080686	0.077329	0.0325161	0.0022465	0.248917
Output tariff	1,205,220	0.1145571	0.097783	0.0646913	0	0.5007753
Consumption good tariff	1,200,683	0.116304	0.1102924	0.0444109	0.0022317	0.4994711
Uncertainty	1,205,220	4.889247	4.905275	0.1405551	4.60517	5.23289
SOE share (2000)	1,205,220	0.2404282	0.2074074	0.1514963	0	1
FDI liberalization	1,205,220	0.2212418	0	0.4150831	0	1
Firm size	1,204,789	9.98199	9.827524	1.300859	0	18.87176
Productivity	1,202,544	6.214546	6.138103	1.095996	−0.2791624	12.61898

Notes: Summary statistics of the main variables in the firm-level data, 2000–2006.

A.3 Baseline OLS without Capital and Output Tariffs

Table A.2: *Effect of Input Tariff Reductions on Firm Labor Share: Stepwise Controls Added.*

	(1)	(2)	(3)
	OLS	OLS	OLS
DepVar:	ls_{it}	ls_{it}	ls_{it}
$\log(1 + \tau_{jt}^M)$	-0.598*** (0.214)	-0.588*** (0.215)	-0.473** (0.212)
$\log(1 + \tau_{jt}^K)$		-0.207 (0.236)	-0.214 (0.235)
$\log(1 + \tau_{jt}^Q)$			-0.197 (0.129)
Firm FE	Y	Y	Y
Year FE	Y	Y	Y
R^2	0.588	0.588	0.588
Observations	1,205,704	1,205,220	1,205,220

Notes: OLS estimates of the impact of intermediate-input tariffs on the firm-level labor share, adding capital and output tariffs sequentially. Robust standard errors in parentheses are clustered at the industry level. *, **, *** denote significance at the 10%, 5%, and 1% levels.

A.4 Additional Controls for Concurrent Reforms

Table A.3: *Robustness: Additional Controls for Concurrent Reforms.*

	(1)	(2)	(3)	(4)	(5)
	Uncertainty	Export Tariff	SOE Reform	FDI Lib.	Firm Controls
DepVar:	ls_{it}	ls_{it}	ls_{it}	ls_{it}	ls_{it}
$\log(1 + \tau_{jt}^M)$	-0.614*** (0.208)	-0.669*** (0.210)	-0.683*** (0.213)	-0.639*** (0.217)	-1.770*** (0.402)
$Uncertainty_{it}$	Y	Y	Y	Y	Y
Export Tariff	N	Y	Y	Y	Y
SOE_Reform_{it}	N	N	Y	Y	Y
FDI_lib_{it}	N	N	N	Y	Y
Firm Controls	N	N	N	N	Y
Controls	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Observations	1,205,220	1,205,220	1,205,220	1,205,220	1,103,910

Notes: All specifications use IV estimation, instrumenting the input tariff with the interaction of the 2001 pre-WTO tariff and a post-2001 indicator. Column (1) adds trade policy uncertainty ($Uncertainty_{it} = col2_i \times Post01_t$). Column (2) adds the industry-level export tariff. Column (3) adds each industry's SOE share interacted with year dummies. Column (4) adds $FDI_lib_{it} = FDI_i \times Post02_t$. Column (5) adds firm size and productivity, which are bad controls; we report it for transparency.

A.5 Alternative Specifications

Table A.4: *Robustness: Sample, Fixed Effects, and Tariff Construction.*

	(1)	(2)	(3)	(4)	(5)
	Balanced	Pref-Year	Winsor	Consump.	IO-Based
	Panel	FE	5%	Tariff	Tariff
DepVar:	ls_{it}	ls_{it}	ls_{it}^{tr}	ls_{it}	ls_{it}
$\log(1 + \tau_{jt}^M)$	-1.025*** (0.333)	-0.484** (0.207)	-0.537** (0.211)	-0.683*** (0.250)	-0.913*** (0.292)
$\log(1 + \tau_{jt}^C)$	N	N	N	Y	N
Pref-year FE	N	Y	N	N	N
Controls	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Observations	263,042	1,205,149	1,068,390	1,200,456	1,206,262

Notes: All specifications are IV. Column (1) restricts to a balanced panel of continuously observed firms. Column (2) adds prefecture-by-year fixed effects. Column (3) winsorizes the top and bottom 5% of the labor share. Column (4) adds the consumption-goods import tariff. Column (5) uses an IO-table-weighted input tariff constructed from China's 2002 IO table. Standard errors clustered at the industry level.

B Mechanism Details

B.1 Decomposition Contributions by Two-Digit Industry

Table B.1: *Decomposition Contributions by Two-Digit Industry.*

ind2	Industry Description	θ_L	Median weights			Mean weights		
			μ	α_M	Total	μ	α_M	Total
		(%)	(%)	(%)	(%)	(%)	(%)	(%)
13	Agri. Food Processing	-13.12	59.35	67.79	114.01	80.61	44.33	111.82
14	Food Manufacturing	-13.12	58.92	50.48	96.27	73.51	25.50	85.89
15	Beverage Mfg.	-13.12	52.31	52.84	92.03	64.55	26.35	77.77
16	Tobacco Mfg.	-13.12	34.90	62.24	84.02	45.72	35.75	68.34
17	Textile	-13.12	66.80	45.16	98.84	79.89	30.65	97.42
18	Textile Apparel & Footwear	-13.12	61.40	35.35	83.62	70.68	24.56	82.11
19	Leather, Fur & Feather	-13.12	61.95	38.59	87.42	73.18	27.82	87.88
20	Timber & Wood Processing	-13.12	59.17	57.13	103.18	72.93	40.05	99.87
21	Furniture Mfg.	-13.12	61.73	46.78	95.39	75.47	30.51	92.86
22	Paper & Paper Products	-13.12	64.97	49.60	101.44	81.76	32.35	100.99
23	Printing & Media Reprod.	-13.12	57.54	28.31	72.73	68.40	7.15	62.43
24	Culture, Educ.& Sport Art.	-13.12	63.94	36.63	87.45	73.56	23.30	83.75
25	Petroleum Proc.& Coking	-13.12	61.20	61.73	109.81	84.58	40.49	111.95
26	Raw Chemical Materials	-13.12	62.32	57.64	106.84	79.99	37.73	104.60
27	Medicines & Pharma.	-13.12	52.41	52.27	91.56	65.04	31.30	83.21
28	Chemical Fibers	-13.12	73.84	52.40	113.13	96.22	28.89	112.00
29	Rubber Products	-13.12	59.52	46.09	92.50	71.88	30.12	88.88
30	Plastic Products	-13.12	66.55	47.80	101.23	82.06	31.44	100.37
31	Non-Metallic Mineral Prod.	-13.12	56.96	40.81	84.65	68.10	20.93	75.91
32	Ferrous Metals Smelting	-13.12	66.88	63.95	117.72	88.38	43.86	119.12
33	Non-Ferrous Metals Smelting	-13.12	70.16	65.02	122.06	95.32	45.66	127.85
34	Metal Products	-13.12	65.59	49.02	101.48	80.29	33.96	101.13
35	General Purpose Machinery	-13.12	61.73	47.07	95.68	73.79	31.24	91.90
36	Special Purpose Machinery	-13.12	59.07	45.66	91.61	70.83	24.26	81.97
37	Transport Equipment	-13.12	62.12	43.31	92.31	76.16	24.82	87.86
39	Electrical Mach. & Equip.	-13.12	67.09	51.07	105.04	82.14	34.54	103.56
40	Comms., Computer & Electr.	-13.12	62.00	41.07	89.94	78.39	22.19	87.46
41	Measuring Instruments	-13.12	59.07	40.80	86.75	72.49	21.50	80.86
42	Artwork & Other Mfg.	-13.12	59.49	41.82	88.19	71.97	29.91	88.77

Notes: Contribution of each margin to the total labor-share response to input-tariff cuts, by two-digit CIC industry. Median and mean weights are within-industry sample medians/means of V/Y and Π/Y applied to the IV coefficients in Table 2 of the main text. The intermediate-input substitution channel is between 40 and 68 percent across industries.

B.2 Wage Regression for Regional Residual Wages

Table B.2: *Wage Regression Used to Construct Regional Residual Wages.*

DepVar:	(1) log w	(2) log w	(3) log w
log(Age)	4.111*** (0.148)	4.265*** (0.148)	3.869*** (0.143)
log(Age) ²	−0.576*** (0.022)	−0.594*** (0.022)	−0.542*** (0.021)
Education: Literacy class	0.169*** (0.023)	0.061 (0.040)	0.046 (0.038)
Education: Primary school	0.338*** (0.022)	0.214*** (0.040)	0.187*** (0.038)
Education: Junior high school	0.601*** (0.023)	0.467*** (0.040)	0.395*** (0.038)
Education: Senior high school	0.889*** (0.024)	0.756*** (0.041)	0.634*** (0.039)
Education: Junior college	1.351*** (0.027)	1.205*** (0.043)	1.021*** (0.041)
Education: Undergraduate and above	1.639*** (0.060)	1.490*** (0.069)	1.310*** (0.064)
Demographics dummies	No	Yes	Yes
Occupation dummies	No	No	Yes
Observations	60,744	60,744	60,744
R^2	0.268	0.277	0.328

Notes: The dependent variable is the log hourly wage. The sample is individuals with urban *Hukou* and weekly hours between 5 and 98 in the 2005 0.1% population census. The omitted education category is “No schooling.” Demographics include sex, marital status, ethnicity, and literacy. Column (3) is the preferred specification used to compute prefecture-level residual wages w_c . Robust standard errors in parentheses.

B.3 Estimated Elasticity of Substitution by Industry

Table B.3: *Estimated Elasticity of Substitution σ_{HM} by Industry.*

Ind.	Description	$\hat{\sigma}_{HM}$	SE
13	Agri. Food Processing	0.996	(0.111)
14	Food Manufacturing	0.727	(0.119)
15	Beverage Mfg.	0.684	(0.169)
17	Textile	0.894	(0.131)
18	Textile Apparel & Footwear	0.699	(0.161)
19	Leather, Fur & Feather	1.046	(0.172)
20	Timber & Wood Processing	1.000	(0.199)
21	Furniture Mfg.	0.901	(0.117)
22	Paper & Paper Products	1.116	(0.116)
23	Printing & Media Reprod.	0.597	(0.161)
24	Culture, Educ. & Sport Art.	0.832	(0.193)
25	Petroleum Proc. & Coking	0.638	(0.215)
26	Raw Chemical Materials	0.790	(0.086)
27	Medicines & Pharma.	0.546	(0.132)
28	Chemical Fibers	1.219	(0.272)
29	Rubber Products	0.790	(0.137)
30	Plastic Products	0.866	(0.086)
31	Non-Metallic Mineral Prod.	0.812	(0.121)
32	Ferrous Metals Smelting	0.822	(0.148)
33	Non-Ferrous Metals Smelting	1.010	(0.178)
34	Metal Products	0.900	(0.091)
35	General Purpose Machinery	0.777	(0.097)
36	Special Purpose Machinery	0.779	(0.089)
37	Transport Equipment	0.627	(0.081)
39	Electrical Mach. & Equip.	0.809	(0.105)
40	Comms., Computer & Electr.	0.718	(0.113)
41	Measuring Instruments	0.822	(0.167)
42	Artwork & Other Mfg.	0.619	(0.198)

Notes: Industry codes follow the two-digit Chinese Industrial Classification (CIC). Estimation follows [Oberfield and Raval \(2021\)](#); for each industry we estimate the regression in Section 5.3 of the main text. Standard errors in parentheses are clustered at the region level. The pooled (employment-weighted) estimate across industries is $\hat{\sigma}_{HM} \approx 0.85$.

C Global Evidence

C.1 Joint Trends in the Labor Share and Trade

Figure C.1 plots the global trends in the labor share and the trade-to-GDP ratio that motivate the analysis in the main text.

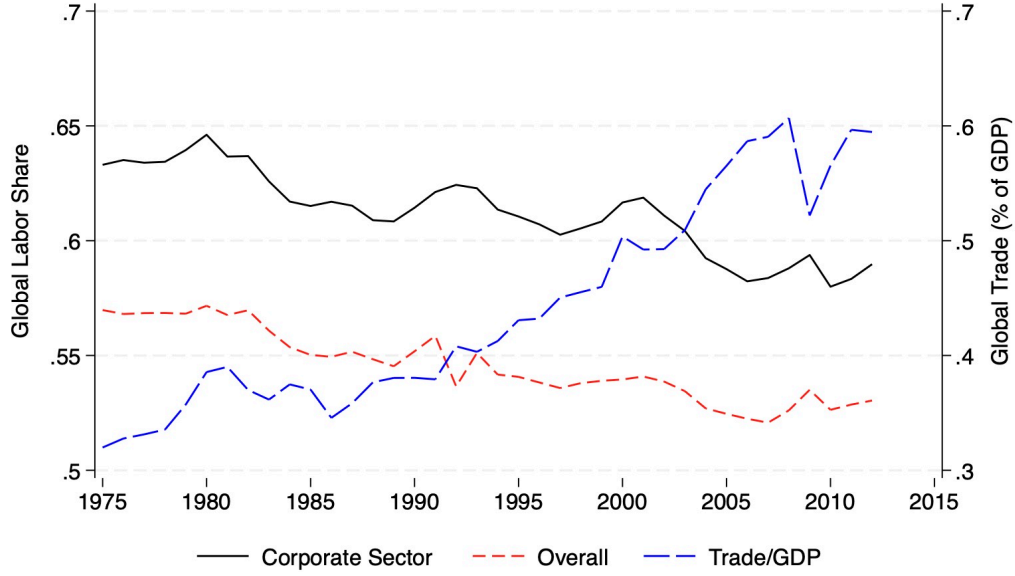


Figure C.1: Declining Labor Share and Growing International Trade

Notes: Global trends in the labor share (left axis) and the trade-to-GDP ratio (right axis), 1975–2012. The labor share series replicates Figure 1 of [Karabarbounis and Neiman \(2014\)](#), who plot the year fixed effects from regressions of corporate and overall labor share that also include country fixed effects. The global trade-to-GDP series is from the World Bank.

C.2 Data Construction for the Country–Industry–Year Sample

Table 3 of the main text reports the baseline global regression on the country–industry–year panel of labor shares constructed from the Socio-Economic Accounts of the World Input–Output Database, covering 43 countries and 56 industries (ISIC Rev.4) over 2000–2014. Country–industry–year intermediate tariffs are constructed by mapping HS6 tariff rates from the World Integrated Trade Solution to ISIC Rev.4 industries using the official United Nations concordance tables, classifying products as intermediate, capital, or consumption goods following the U.S. Bureau of Economic Analysis convention, and weighting by import shares within each industry. Capital and output tariffs are constructed analogously. The estimating equation mirrors equation (3) of the main text at the country–industry–year level, with country, industry, and year fixed effects.

C.3 Robustness with Alternative Fixed Effects and Tariff Construction

Table C.1: *Effect of Input Tariffs on Industry Labor Share: Robustness.*

	(1)	(2)	(3)	(4)	(5)	(6)
DepVar:	ls_{it}	ls_{it}	ls_{it}	ls_{it}	ls_{it}	ls_{it}
$\log(1 + \tau_{jt}^M)$	-0.0151* (0.00781)	-0.0156** (0.00791)	-0.0268*** (0.00806)	-0.0310** (0.0104)	-0.0275*** (0.00813)	-0.0375*** (0.00844)
$\log(1 + \tau_{jt}^K)$		0.00828 (0.0208)	0.00925 (0.0208)	0.145*** (0.0235)	-0.00608 (0.0212)	-0.0161 (0.0216)
$\log(1 + \tau_{jt}^Q)$			0.00128 (0.000181)	0.00137*** (0.000205)	0.00131*** (0.000183)	0.000937*** (0.000184)
Country FE	Y	Y	Y	N	Y	N
Industry FE	Y	Y	Y	N	N	Y
Year FE	Y	Y	Y	Y	N	N
Country-Industry FE	N	N	N	Y	N	N
Country-Year FE	N	N	N	N	N	Y
Industry-Year FE	N	N	N	N	Y	N
R^2	0.459	0.459	0.460	0.189	0.469	0.485
Observations	15,887	15,887	15,887	15,887	15,887	15,887

Notes: Country–industry–year observations from 2000 to 2014. Columns (1)–(3) use tariff rates aggregated with constant input-output coefficients. Columns (4)–(6) use the main-text tariffs with alternative fixed-effects structures: country-by-industry plus year (Col. 4), country plus industry-year (Col. 5), and industry plus country-year (Col. 6). Robust standard errors in parentheses are clustered at the industry level.

D Theory Derivations

This appendix reproduces the derivations underlying the decomposition in equations (1) and (2) of the main text and presents a Melitz–Ottaviano model with imported intermediate inputs that microfound the negative markup response to input-tariff cuts documented in Table 2.

D.1 Derivation of the Labor-Share Identity

Let gross output be priced at a markup over input costs, $V = \mu(RK + WL + qM)$, with constant returns to scale. Define value added as $Y = V - qM$. Substituting,

$$\begin{aligned}
Y &= \mu(RK + WL + qM) - qM \\
&= (RK + WL) + (\mu - 1)(RK + WL) + (\mu - 1)qM.
\end{aligned}$$

The three terms are primary input cost, profits on primary input, and profits on intermediates. With $\alpha_M \equiv qM/(RK + WL + qM)$ and $\theta_L \equiv WL/(RK + WL)$, the labor share is

$$s_L = \frac{WL}{Y} = \frac{\theta_L(RK + WL)}{Y}.$$

From the value-added decomposition above, $Y/(RK + WL) = 1 + (\mu - 1)/(1 - \alpha_M)$, which delivers equation (1) of the main text,

$$s_L = \frac{\theta_L}{1 + \frac{\mu - 1}{1 - \alpha_M}}.$$

Totally differentiating and using $V/Y = 1 + (\mu - 1)(RK + WL + qM)/Y$ and $\Pi/Y = (\mu - 1)(RK + WL + qM)/Y$ delivers equation (2) of the main text,

$$\hat{s}_L = \hat{\theta}_L - \left(\frac{V}{Y}\right)\hat{\mu} + \left(\frac{\Pi}{Y}\right)\widehat{(1 - \alpha_M)}.$$

D.2 Tariff Pass-Through to the Relative Input Price

Under the Cobb–Douglas composite $M = M_D^\beta M_X^{1-\beta}$ with $\beta \in (0, 1)$, gross domestic tariff τ , world price P_X , and a delivered import price τP_X , the unit cost of the intermediate bundle is

$$q = B P_D^\beta (\tau P_X)^{1-\beta}, \quad B \equiv \beta^{-\beta} (1 - \beta)^{\beta-1}.$$

Holding domestic input prices fixed,

$$\frac{\partial \ln(q/c_H)}{\partial \ln \tau} = 1 - \beta > 0,$$

so a tariff cut lowers the relative price of intermediates. Combining with the CES relation $\partial \ln[\alpha_M/(1 - \alpha_M)]/\partial \ln(q/c_H) = 1 - \sigma_{HM}$ gives

$$\left. \frac{\partial \ln[\alpha_M/(1 - \alpha_M)]}{\partial \ln \tau} \right|_Q = (1 - \sigma_{HM})(1 - \beta),$$

which is the comparative-static result reported in the main text.

D.3 A Melitz–Ottaviano Model with Imported Intermediate Inputs

We provide a closed-form derivation showing that, under quasi-linear quadratic demand, free entry, and a Pareto productivity distribution, an input-tariff reduction lowers the average markup in equilibrium. This microfounds the positive coefficient on $\log(1 + \tau^M)$ in Column (3) of Table 2.

Consumer demand. The representative consumer has quasi-linear quadratic utility

$$U = q_0 + \alpha \int_0^N q(i) di - \frac{\eta}{2} \int_0^N q(i)^2 di - \frac{\gamma}{2} \left(\int_0^N q(i) di \right)^2,$$

yielding inverse demand $p(i) = \alpha - \eta q(i) - \gamma Q$ with $Q \equiv \int_0^N q(j) dj$.

Production. Firms differ in productivity ϕ and produce with Cobb–Douglas technology over labor ℓ and imported intermediate m , $q = \phi \ell^{1-\beta} m^\beta$. Normalizing $w = 1$ and writing the tariff-inclusive intermediate price as $p_m = (1 + \tau)p_f$, marginal cost is

$$c(\phi, \tau) = \frac{A(\tau)}{\phi}, \quad A(\tau) \equiv \left(\frac{1}{1-\beta}\right)^{1-\beta} \left(\frac{p_f}{\beta}\right)^\beta (1 + \tau)^\beta.$$

$A(\tau)$ is increasing in τ .

Firm optimization. Defining the choke price $p_{\max} \equiv \alpha - \gamma Q$, profit maximization yields

$$q(\phi, \tau) = \frac{1}{2\eta}(p_{\max} - c(\phi, \tau)), \quad p(\phi, \tau) = \frac{1}{2}(p_{\max} + c(\phi, \tau)),$$

$$m(\phi, \tau) = \frac{1}{2}(p_{\max} - c(\phi, \tau)), \quad \pi_{op}(\phi, \tau) = \frac{1}{4\eta}(p_{\max} - c(\phi, \tau))^2.$$

Firms produce iff $c(\phi, \tau) < p_{\max}$.

Equilibrium. The zero-profit cutoff productivity ϕ_D satisfies $\pi_{op}(\phi_D, \tau) = f_D$, which gives $p_{\max} = A(\tau)/\phi_D + 2\sqrt{\eta f_D}$. The free-entry condition with a Pareto distribution (shape $k > 2$, lower bound $\phi_{\min}$) becomes

$$\left[\frac{A(\tau)^2}{2\eta(k+1)(k+2)\phi_D^2} + \frac{A(\tau)\sqrt{f_D/\eta}}{(k+1)\phi_D} \right] \left(\frac{\phi_{\min}}{\phi_D} \right)^k = f_E.$$

By the implicit function theorem, since the left-hand side is decreasing in ϕ_D and increasing in $A(\tau)$, $d\phi_D/d\tau > 0$.

Markup response. Using $\mathbb{E}[1/\phi \mid \phi \geq \phi_D] = k/[(k+1)\phi_D]$, the average markup of surviving firms is

$$\overline{m}(\tau) = \frac{c_D(\tau)}{2(k+1)} + \sqrt{\eta f_D}, \quad c_D(\tau) \equiv \frac{A(\tau)}{\phi_D(\tau)}.$$

Rewriting the free-entry condition in c_D shows $dc_D/d\tau > 0$, hence

$$\frac{d\overline{m}}{d\tau} = \frac{1}{2(k+1)} \frac{dc_D}{d\tau} > 0.$$

A reduction in the intermediate-input tariff therefore lowers the average markup. The intuition is that under linear demand, the partial-equilibrium force from cheaper inputs raises individual markups; this however raises profits, induces entry, and the pro-competitive general-equilibrium response dominates. This is exactly the sign of $\log(1+\tau^M)$ on μ in Column (3) of Table 2.

D.4 Aggregate vs. Firm-Level Elasticities

Oberfield and Raval (2021) show that the aggregate elasticity of substitution between primary and intermediate inputs may differ substantially from its firm-level counterpart, generically rising with the level of aggregation. The firm-level estimate $\hat{\sigma}_{HM} \approx 0.85 < 1$ documented in Section 5.3 therefore does *not* imply that the aggregate elasticity is below one. If the aggregate elasticity exceeds one, an aggregate decline in the relative price of intermediate inputs would raise α_M at the aggregate level, contributing to the secular decline in the aggregate labor share even as firm-level liberalization raises individual labor shares. Reconciling firm-level and aggregate responses through a structural mapping is left for future work.